Day 31 – Dockerfile: Build Your Own Images

Task 1: Your First Dockerfile

Step 1: Create Project Folder

```
None
mkdir my-first-image
cd my-first-image
```

Step 2: Create Dockerfile

```
None
nano Dockerfile
```

Paste:

```
None
FROM ubuntu

RUN apt-get update && apt-get install -y curl

CMD ["echo", "Hello from my custom image!"]
```

My-first-image project1

FROM ubuntu

WORKDIR /app

RUN apt-get update && apt-get install -y curl

CMD ["echo", "Hello from my custom image!"]

Save and exit.

Step 3: Build Image

```
None
docker build -t my-ubuntu:v1 .
```

Explanation:

- `docker build` → build image
- `-t` → tag image
- `my-ubuntu:v1` → image name and version
- `.` → current directory (build context)

```
ubuntu@ip-172-31-81-141:~/my-first-image$ docker build -t my-first-image .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/4 : FROM ubuntu
latest: Pulling from library/ubuntu
1c24335ddd46: Pulling fs layer
6f5c5aa4e145: Pulling fs layer
1c24335ddd46: Download complete
9bcf140d7f0f: Download complete
6f5c5aa4e145: Download complete
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
 ---> f3d28607ddd7
Step 2/4 : WORKDIR /app
 ---> Running in 3ba8ccad4f53
 ---> Removed intermediate container 3ba8ccad4f53
 ---> fd4c92e8489e
Step 3/4 : RUN apt-get update && apt-get install -y curl
 ---> Running in 967f57f02454
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [250 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [260 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [142 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [3584 B]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [160 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [279 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [250 kB]
Fetched 24.4 MB in 2s (14.5 MB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
```

Step 4: Verify Image

None

```
docker images
```

```
ubuntu@ip-172-31-81-141:~/my-first-image$ docker images

IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
devboard:latest          c1c4554d2bb5       538MB          103MB    U
flaskapp:latest          1687f15e61fe      1.64GB          421MB    U
my-first-image:latest    f4b69f4e237d       254MB         77.1MB    U
node:20-alpine           fb4cd12c85ee       194MB         48.8MB
python:3.14              6f473f84b09f      1.63GB          432MB
ubuntu:latest            f3d28607ddd7       160MB         45.3MB
ubuntu@ip-172-31-81-141:~/my-first-image$
```

Output:

```
None
REPOSITORY    TAG
my-ubuntu     v1
```

Step 5: Run Container

```
None
docker run my-ubuntu:v1
```

Output:

```
None
Hello from my custom image!
```

✅ Task 1 Complete

```
ubuntu@ip-172-31-81-141:~/my-first-image$ docker run my-first-image
Hello from my custom image!
ubuntu@ip-172-31-81-141:~/my-first-image$
```

Task 2: Dockerfile Instructions

—----------**chatgpt**—----------

If you want to host an `index.html` page using Nginx in Docker, create these two files:

Dockerfile

```
None
FROM nginx:alpine

COPY index.html /usr/share/nginx/html/index.html
```

```
EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

---

index.html

None

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width,
initial-scale=1.0">
   <title>My Docker Nginx Website</title>
   <style>
       body {
           font-family: Arial, sans-serif;
           text-align: center;
           margin-top: 100px;
           background-color: #f4f4f4;
       }
       .container {
           background: white;
           padding: 30px;
           width: 60%;
           margin: auto;
           border-radius: 10px;
           box-shadow: 0 0 10px gray;
       }
       h1 {
           color: #0078d7;
       }
   </style>
</head>
<body>
```

```
    <div class="container">
        <h1>Welcome to Docker + Nginx!</h1>
        <p>This website is running inside a Docker container.</p>
        <p>Build Date: 17 June 2026</p>
        <button onclick="showMessage()">Click Me</button>
        <p id="message"></p>
    </div>

    <script>
        function showMessage() {
            document.getElementById("message").innerHTML =
                "Congratulations! Your Nginx container is serving
this page.";
        }
    </script>
</body>
</html>
```

Build the image

```
None
docker build -t my-nginx-site .
```

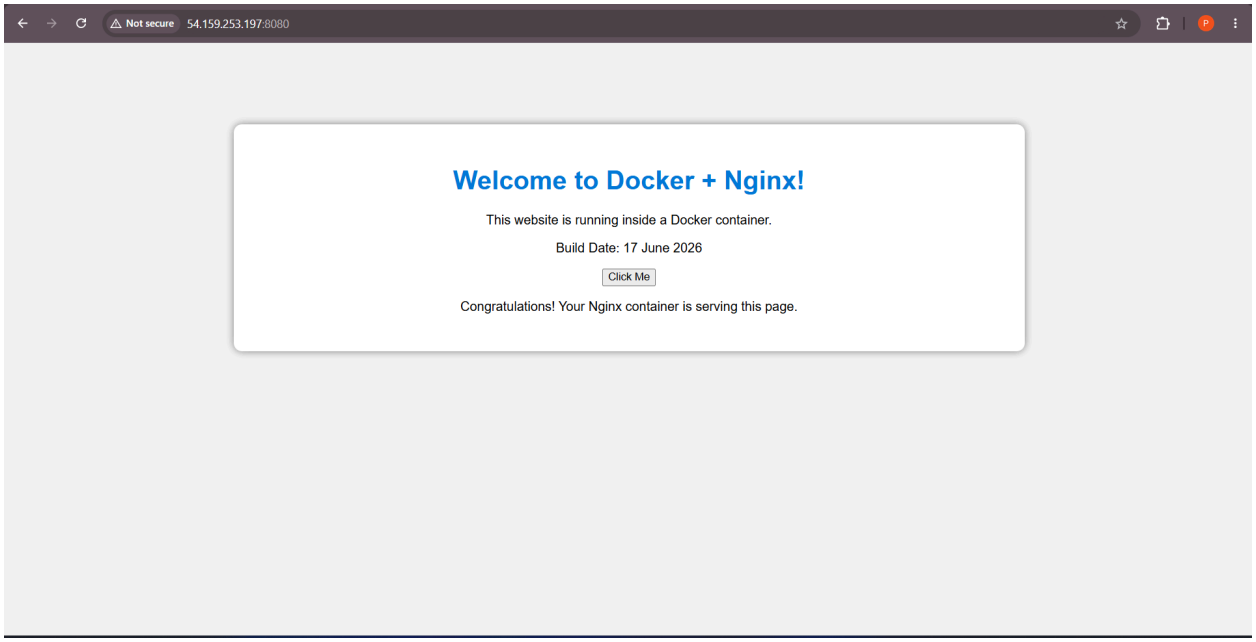
Run the container

```
None
docker run -d -p 8080:80 --name nginx-web my-nginx-site
```

Access the website

Open:

```
None
http://localhost:8080
```



You should see your custom HTML page instead of the default Nginx welcome page.

Flask-app-ecs-project2—-------Dockerfile

```
FROM python:3.14

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

CMD ["python", "run.py"]
```

Index.html project3—----------Dockerfile

```
FROM nginx:alpine

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

```
ubuntu@ip-172-31-81-141:~/indexexample$ ls
Dockerfile  index.html
ubuntu@ip-172-31-81-141:~/indexexample$ cat Dockerfile
FROM nginx:alpine

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Meaning of Each Instruction

FROM

Base image.

```
None
FROM nginx:alpine
```

RUN

Runs commands while building image.

```
None
RUN echo "Build Successful"
```

COPY

Copies files from host to image.

```
None
COPY index.html .
```

WORKDIR

Sets working directory.

```
None
WORKDIR /usr/share/nginx/html
```

EXPOSE

Documents port.

```
None
EXPOSE 80
```

CMD

Default command.

```
None
CMD ["nginx","-g","daemon off;"]
```

Task 3: CMD vs ENTRYPOINT

CMD Example

Dockerfile:

```
None
FROM ubuntu
```

```
CMD ["echo", "hello"]
```

Build:

```
None
docker build -t cmd-demo .
```

Run:

```
None
docker run cmd-demo
```

Output:

```
None
hello
```

Override CMD:

```
None
docker run cmd-demo date
```

Output:

```
None
Sun Jun 15 ...
```

CMD gets replaced.

ENTRYPOINT Example

Dockerfile:

```
None
FROM ubuntu

ENTRYPOINT ["echo"]
```

Build:

```
None
docker build -t entry-demo .
```

Run:

```
None
docker run entry-demo hello
```

Output:

```
None
hello
```

Run again:

```
None
docker run entry-demo docker
```

Output:

```
None
docker
```

Arguments are appended.

CMD vs ENTRYPOINT

CMD	ENTRYPOINT
Default command	Fixed command
Easily overridden	Not overridden easily
Used for optional defaults	Used for mandatory executable

Use CMD

```
None
CMD ["python","app.py"]
```

When user may want another command.

Use ENTRYPOINT

```
None
ENTRYPOINT ["python"]
```

When image should always run Python.

Task 4: Build a Simple Web App

Step 1: Create Folder

```
None
mkdir my-website
cd my-website
```

Step 2: Create HTML

```
None
nano index.html
```

Paste:

```
None
<!DOCTYPE html>
<html>
<head>
<title>Docker Website</title>
</head>
<body>
<h1>Hello From Docker!</h1>
</body>
</html>
```

Step 3: Create Dockerfile

```
None
FROM nginx:alpine

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
```

Step 4: Build

```
None
docker build -t my-website:v1 .
```

Step 5: Run

```
None
docker run -d -p 8080:80 --name website my-website:v1
```

Step 6: Verify

Open:

```
None
http://localhost:8080
```

or

```
None
http://EC2-PUBLIC-IP:8080
```

You should see:

```
None
Hello From Docker!
```

✅ Task 4 Complete

Task 5: .dockerignore

Create file:

```
None
nano .dockerignore
```

Paste:

```
None
node_modules
.git
*.md
.env
```

Verify

Build image:

```
None
docker build -t ignore-demo .
```

Check build context size:

```
None
Sending build context to Docker daemon
```

Size becomes smaller because ignored files are excluded.

Task 6: Build Optimization

Example Dockerfile

```
None
FROM ubuntu

COPY . .

RUN apt-get update
```

```
CMD ["bash"]
```

Build:

```
None
docker build -t cache-demo .
```

Modify:

```
None
README.md
```

Build again:

```
None
docker build -t cache-demo .
```

Docker rebuilds from COPY layer onward.

Better Dockerfile

```
None
FROM ubuntu

RUN apt-get update

COPY . .

CMD ["bash"]
```

Now package installation layer is cached.

Only COPY layer rebuilds.

Why Layer Order Matters

Docker builds images in layers:

```
None
Layer 1 → FROM
Layer 2 → RUN
Layer 3 → COPY
Layer 4 → CMD
```

If Layer 3 changes:

```
None
Layer 1 = Cached
Layer 2 = Cached
Layer 3 = Rebuilt
Layer 4 = Rebuilt
```

Best Practice:

```
None
Put stable commands first
Put frequently changing files last
```

This makes builds much faster.

Important Commands Cheat Sheet

```
None
docker build -t image:tag .

docker images
```

```
docker run image:tag

docker run -d -p 8080:80 image:tag

docker ps

docker ps -a

docker stop container_name

docker rm container_name

docker rmi image_name

docker logs container_name
```

Devboard project4—----------Dockerfile

From node:20-alpine

WORKDIR /app

COPY package*.json ./

RUN npm ci

COPY . .

RUN npm run build

COPY . .

EXPOSE 5173

CMD ["npm","run","preview","--","--host","0.0.0.0","--port","5173"]

This is enough to complete **Day 31 Dockerfile Challenge** and create your `day-31-dockerfile.md` for GitHub.

If you want to delete **all containers** and **all images**, run these commands in order.

1. Stop all running containers

```
None
docker stop $(docker ps -q)
```

2. Delete all containers

```
None
docker rm $(docker ps -aq)
```

3. Delete all images

```
None
docker rmi $(docker images -q)
```

If Docker says an image is being used by a container, make sure you've removed all containers first (steps 1 and 2).

4. Verify everything is deleted

```
None
docker ps -a
docker images
```

Both should show no containers/images.